# TSHR and IGF1R in Graves disease and thyroid eye disease: genetic evidence from blood expression

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## Abstract

**Objective:** Thyroid eye disease (TED) can accompany Graves disease (GD), an autoimmune cause of hyperthyroidism. We compared genetic evidence linking blood *TSHR* and *IGF1R* expression with susceptibility.

**Methods:** We screened 2,544 druggable genes using Mendelian randomization (MR). Disease outcomes were Biobank Japan (BBJ) GD, UK Biobank (UKB) hyperthyroidism and overlapping FinnGen GD, Graves ophthalmopathy (GO) and strict GO endpoints. Colocalization tested 24 genes for a shared causal variant. Tissue eQTL and orbital RNA-sequencing (four TED, one control) were supplementary.

**Results:** Thirteen genes passed BBJ discovery ($P < 0.05/2{,}544$). Higher genetically proxied *TSHR* expression was associated with lower GD odds in BBJ (odds ratio [OR] per approximately one standard deviation 0.12, 95% confidence interval [CI] 0.07–0.21; $P = 1.09\times10^{-14}$) and FinnGen (OR 0.08, 95% CI 0.05–0.12; $P = 3.19\times10^{-32}$), concordantly across all five datasets. Shared-variant probability exceeded 0.80 in FinnGen GO and, prior-sensitively, BBJ and strict GO, but UKB (0.226) and FinnGen GD (0.264) favored distinct variants. *IGF1R* ORs exceeded 1 but reached $P < 0.05$ only in BBJ (OR 1.56, 95% CI 1.07–2.28; $P = 0.0212$) and UKB ($P = 0.0117$); the most precise estimate, FinnGen GD, was OR 1.07 (95% CI 0.86–1.33). Single-variant tissue estimates added no independent evidence. Mean orbital *TSHR* and *IGF1R* transcript levels exceeded the control.

**Conclusions:** Association evidence was stronger for *TSHR* than *IGF1R*, but absent colocalization in UKB and FinnGen GD leaves expression-mediated causality unproven. Population-controlled data cannot isolate TED susceptibility within GD; weaker *IGF1R* evidence does not argue against IGF-1R–targeted treatment.

**Keywords:** Graves disease; thyroid eye disease; *TSHR*; *IGF1R*; Mendelian randomization; colocalization

## Introduction

Thyroid eye disease (TED) can cause proptosis, diplopia and visual impairment in patients with Graves disease (GD) [1, 2]. Teprotumumab, which blocks the insulin-like growth factor 1 receptor (IGF-1R), improves clinical outcomes in active TED [3]. Its efficacy raises a related but distinct question: does inherited variation influencing *IGF1R* expression contribute to disease susceptibility in the same way as variation at *TSHR*, the established GD autoantigen and susceptibility locus?

TSHR and IGF-1R interact in experimental models of orbital disease [4, 5, 6]. However, treatment response, inherited gene regulation and gene expression in inflamed tissue address different biological questions. A receptor may be a useful treatment target without equally strong genetic evidence linking its expression to disease onset. Distinguishing these forms of evidence helps clinicians interpret genetic studies of therapeutic targets.

Mendelian randomization (MR) uses variants associated with an exposure to investigate its relationship with the disease phenotype being studied, termed the outcome [7]. Here, the exposure is blood gene expression and the outcomes are disease diagnoses, not responses to treatment. Because alleles are allocated at conception, MR is less prone than ordinary observational studies to confounding and reverse causation, but it supports causal interpretation only when its instrumental-variable assumptions hold. Colocalization asks whether expression and disease associations in a genomic region are consistent with the same causal variant [8]. Distinct nearby variants can appear connected because they are inherited together. Thus, an MR association may be highly significant even when the regional signals do not support a shared variant. MR reports odds ratios (ORs), confidence intervals (CIs) and *P* values; colocalization reports model-based posterior probabilities, which are not *P* values.

Previous genetic studies have already implicated *TSHR* in GD and Graves ophthalmopathy. Ji and colleagues integrated blood methylation, expression and protein associations with GD, reporting *TSHR* among the expression-associated genes also supported in a strict FinnGen ophthalmopathy dataset [9]. Other MR work has examined inflammatory proteins in ophthalmopathy [10], and recent integrative research has investigated *TSHR* across genetic and transcriptomic data in GD and ophthalmopathy [11]. These findings establish a basis for comparing clinically relevant receptors, but do not determine whether evidence for their blood expression is consistently supported across disease definitions under the same analysis framework. In particular, limited evidence for *IGF1R* could reflect imprecision or dependence on particular instruments, while a strong MR association for *TSHR* does not by itself establish that expression and disease share a causal variant. The remaining question is how robustly these two expression traits are associated with GD and related phenotypes, and how far those associations support biological interpretation.

To address this question, we compared genetically proxied blood expression of *TSHR* and *IGF1R* — expression predicted from inherited variants rather than measured directly — within a screen of genes encoding potentially druggable proteins, using *CTLA4* and other associated genes as context. We used Biobank Japan (BBJ) GD for discovery, compared the original findings with UK Biobank (UKB) hyperthyroidism and FinnGen ophthalmopathy, and subsequently added FinnGen GD and a stricter ophthalmopathy definition. We evaluated effect direction, precision and statistical support across these datasets, then examined dependence on individual instruments, allele frequencies and colocalization assumptions. A later fixed-panel tissue extension evaluated whether these patterns extended beyond the original blood instruments. Our aim was to characterize the consistency and limits of expression-related genetic evidence for the two receptors. The analyses distinguish GD from broader hyperthyroidism and population-controlled ophthalmopathy; they do not directly

estimate TED susceptibility among patients who already have GD or predict treatment response. A small in-house orbital RNA-sequencing series (four TED samples and one control) is reported only as descriptive context.

## Methods

### Study design and disease datasets

This two-sample MR study used public association summary statistics and followed STROBE-MR reporting guidance [12]. Exposure data were whole-blood expression quantitative trait locus (eQTL; a variant associated with gene expression) statistics from predominantly European eQTLGen participants [13]. The original outcomes were BBJ GD, UKB self-reported hyperthyroidism or thyrotoxicosis, and Graves ophthalmopathy in FinnGen release 12 (FinnGen ophthalmopathy) [14, 15, 16]. We later added two FinnGen endpoints: E4_GRAVES_STRICT (FinnGen GD) and E4_GRAVES_OPHT_STRICT (strict ophthalmopathy). Table 1 gives case and control counts. UKB ancestry was not established as exclusively European.

BBJ was the discovery dataset, FinnGen GD a same-disease comparison, UKB a broader-phenotype comparison, FinnGen ophthalmopathy a TED-enriched comparison and strict ophthalmopathy a definition sensitivity analysis. The three FinnGen endpoints share participants and were neither treated as independent replications nor meta-analyzed. Their controls were all other participants, not GD patients without TED. The FinnGen extensions, expanded candidate assessment and instrument diagnostics were post hoc; the study was not prospectively registered. Analyses used R 4.3.3, TwoSampleMR 0.7.4, coloc 5.2.3 and PLINK 1.9, with independent numerical verification in Python.

**Expression instruments and MR**

For 4,462 potentially druggable genes [17], instruments (variants used to represent a gene's expression) were blood eQTLs with $P < 5\times10^{-8}$ within 1 Mb of the gene (cis-eQTLs). PLINK clumping kept only uncorrelated variants (linkage disequilibrium [LD] $r^2 < 0.001$ within 10 Mb, in 503 European participants from 1000 Genomes Phase 3) [18, 19]. This gave 6,135 single-nucleotide polymorphism (SNP)–gene pairs for 2,544 genes (minimum F statistic 29.7; $F > 10$ is the usual threshold for adequate instrument strength), and the extended major histocompatibility complex region was not excluded. Variant effects on expression, in standard deviation units, were derived from eQTLGen Z statistics, sample sizes and 1000 Genomes European allele frequencies [20]. TwoSampleMR aligned exposure and disease effects to the same allele, excluding unavailable and strand-ambiguous (palindromic) variants. UKB linear-model estimates were rescaled using the case proportion to approximate the log-odds scale of BBJ and FinnGen (Supplementary Methods).

For each gene and dataset, MR used the Wald ratio (disease effect divided by expression effect) for one instrument and, for several, multiplicative random-effects inverse-variance weighting (IVW), a precision-weighted average [21]. Estimates are ORs per approximately one standard deviation higher genetically proxied blood expression, with 95% CIs and two-sided *P* values. The BBJ screen threshold was Bonferroni-corrected for all 2,544 instrumented genes, estimable or not ($P < 0.05/2{,}544 = 1.965\times10^{-5}$); it was applied descriptively to the other datasets and in an exploratory full FinnGen GD screen. The 13 BBJ discoveries were assessed at a focused FinnGen GD threshold ($P < 0.05/13$ and the same direction). Nominal (uncorrected) $P < 0.05$ was reported descriptively. ORs detectable with 80% power were approximated from observed standard errors (Supplementary Methods).

MR assumes that instruments predict expression, are unrelated to confounders of expression and disease, and affect disease only through that gene's expression.

For estimates with several instruments, Cochran's Q test assessed heterogeneity among instruments and, for selected genes, post hoc omission of each instrument in turn (leave-one-out) assessed dependence on single variants. MR-Egger regression, whose intercept tests whether instruments affect disease through other pathways (directional pleiotropy), and weighted-median and weighted-mode estimators [22, 23] were applied only in the original three datasets (Supplementary Dataset, MR_estimators_original). Analyses were repeated with exposure effects reconstructed from eQTLGen cohort allele frequencies; for *C4B* and *HLA-G*, whose FinnGen GD screen result changed, instrument weights and omissions were compared between frequency sources (Supplementary Methods).

In a post hoc audit, rs179252 and rs179247 (the variant examined in published thymic *TSHR* studies) were re-extracted from the original expression and disease files. Disease effects were aligned to the blood expression-increasing allele to give per-allele ORs and recalculate the MR estimates, and the correlation between the two variants' G-allele counts was calculated in European and East Asian 1000 Genomes genotypes. The audit checked allele direction, not thymic mediation (Supplementary Dataset, TSHR_allele_audit).

**Colocalization**

MR associations are the main findings, and colocalization was used to interpret them; the original analysis reported both in parallel, and this emphasis was adopted post hoc without prospective registration. We used coloc.abf, which assumes at most one causal variant per trait, with all available eQTLGen cis-eQTL statistics for each gene, significant or not, matched to each disease dataset [8]. It was applied to 24 genes: the 13 BBJ discoveries, *IGF1R*, the nine additional FinnGen GD screen discoveries and *HLA-G*, added because its MR result changed with the allele-frequency source. Posterior probabilities (PP.H0–PP.H4) were estimated for five hypotheses: neither trait associated (H0), disease only (H1), expression only (H2), distinct causal variants (H3) and one shared causal variant (H4). Prior

probabilities that a variant is associated with disease (p1), expression (p2) or both (p12) were $10^{-4}$, $10^{-4}$ and $10^{-5}$, with p12 also set to $5\times10^{-6}$ and $10^{-6}$ [24]. PP.H4 ≥ 0.80 was a descriptive reporting threshold.

For *TSHR*, colocalization was repeated in narrower regions and in a variant set common to all five datasets, and we checked whether the matched variants included each dataset's strongest regional disease variant (Supplementary Methods).

**Tissue-specific extension**

In a post hoc extension, we assessed *TSHR* and *IGF1R* in a fixed panel of five Genotype-Tissue Expression (GTEx) v8 tissues: thyroid, cultured skin fibroblasts, subcutaneous adipose, visceral adipose and whole blood [25]. The plan was fixed after the blood results were known but before tissue results were retrieved. GTEx donors were predominantly European. Instruments required cis-eQTL $P < 5\times10^{-8}$, F > 10 and European-reference LD $r^2 < 0.001$. Ten planned gene–tissue comparisons against FinnGen GD formed the primary family ($P < 0.005$); 40 comparisons against the other four datasets formed a secondary family ($P < 0.00125$). Unestimable combinations remained in the correction denominators. Effects used GTEx slopes and standard errors for inverse-normal-transformed expression. Tissue colocalization was not performed because complete regional statistics were not obtained.

**Descriptive orbital expression**

In-house RNA sequencing of orbital adipose tissue from four patients with inactive TED and of an upper-eyelid nasal fat pad from one control without thyroid disease was descriptive only (Supplementary Methods). Transcript abundance was displayed without significance testing (Figure S1) and was not used to select genes or validate the genetic findings.

## Results

### Discovery across druggable genes

Of the 2,544 instrumented genes, 2,234 were estimable in BBJ, 2,505 in UKB and 2,480 in each FinnGen endpoint. Thirteen genes passed the BBJ screen threshold ($P < 0.05/2{,}544$), including *TSHR* and *CTLA4* but not *IGF1R* (Figure 1). At this threshold, 14.6% and 35.6% of estimable BBJ genes had 80% power to detect ORs of 1.5 and 2.0, respectively (Supplementary Dataset, Detection_original).

### TSHR and IGF1R across disease datasets

Higher genetically proxied blood *TSHR* expression was associated with lower odds of GD in BBJ (OR 0.12, 95% CI 0.07–0.21; $P = 1.09\times10^{-14}$) and FinnGen (OR 0.08, 95% CI 0.05–0.12; $P = 3.19\times10^{-32}$). ORs were also below 1 for UKB hyperthyroidism and both ophthalmopathy endpoints, and all estimates except strict ophthalmopathy passed the screen threshold (Table 2; Figure 2). Each estimate used the single instrument rs179252. Its expression-increasing G allele had per-allele disease ORs of 0.77–0.82 across datasets and was correlated with rs179247-G (European r = 0.978; East Asian r = 0.996), which also increased blood *TSHR* expression. Allele-aligned recalculation reproduced the MR estimates (Supplementary Dataset, TSHR_allele_audit).

All five *IGF1R* ORs exceeded 1, and none passed the screen threshold. Nominal associations (uncorrected $P < 0.05$) were found only in BBJ GD (OR 1.56, 95% CI 1.07–2.28; $P = 0.0212$) and UKB hyperthyroidism (OR 1.35, 95% CI 1.07–1.70; $P = 0.0117$). The most precise estimate, FinnGen GD, was OR 1.07 (95% CI 0.86–1.33; $P = 0.539$); this comparison had 80% power to detect an OR of 1.36 at $P < 0.05$ (Supplementary Methods). Weighted-median ORs were 1.56 ($P = 0.0288$) in BBJ, 1.42 ($P = 0.00244$) in UKB and 1.46 ($P = 0.124$) in FinnGen ophthalmopathy. In these three datasets, MR-Egger and weighted-mode estimates

had $P \geq 0.32$, and MR-Egger intercept $P$ values were 0.66, 0.56 and 0.43, respectively (Supplementary Dataset, MR_estimators_original).

**Other associated genes**

Four genes passed the screen threshold in both GD datasets: *TSHR*, *CTLA4*, *HLA-DQA2* and *TUBB* (Table 3). *CTLA4* was associated with lower odds, and *HLA-DQA2* and *TUBB*, both in the major histocompatibility complex (MHC) region, with higher odds. *CTLA4* and *TUBB* had direction-consistent nominal associations in all five datasets; *HLA-DQA2* did in four, but its two-instrument UKB estimate was non-significant and below 1 (Supplementary Dataset, MR_primary). At the focused FinnGen GD threshold ($P < 0.05/13 = 0.00385$), seven of the 13 BBJ discoveries had direction-consistent associations: these four genes plus *C4A*, *TNFSF14* and *IFNGR1*. *HLA-A*, which had the smallest BBJ $P$ value, had a non-significant FinnGen GD estimate.

The exploratory full FinnGen GD screen identified 13 genes, nine of them outside the BBJ discovery set (Table S1). Seven of the nine were instrumented in the extended MHC region, and two pairs each shared one instrument.

**Instrument and frequency sensitivity**

Omitting rs13030124 left non-significant *CTLA4* estimates in UKB and all three FinnGen endpoints ($P = 0.323–0.972$; Figure S2). In BBJ and UKB, where *IGF1R* had four instruments, omitting each in turn gave $P > 0.05$ in two of the four estimates; omitting rs2654980 in UKB also lowered the OR from 1.35 to 1.16 (Figure S2). Among the BBJ discoveries, Cochran's Q test in FinnGen GD indicated heterogeneity among instruments for *C4A* ($P = 1.27\times10^{-12}$), *TUBB* ($P = 3.67\times10^{-5}$), *HLA-A* ($P = 0.0387$) and *CTLA4* (Table 2). Omitting rs149543464 (*TUBB*) or rs1269852 (*C4A*) raised the FinnGen GD $P$ value to 0.0262 and 0.0471, respectively (Supplementary Dataset, LOO_new_fixed14).

With eQTLGen cohort allele frequencies, all 13 BBJ discoveries still passed the screen threshold, *TSHR P* values were unchanged and no *IGF1R P* value crossed 0.05 (Supplementary Dataset, MR_original_freq and MR_new_freq). Screen classification changed only in FinnGen GD: *C4B* no longer passed, and *HLA-G* newly passed; the IVW weight of rs150254595 in its estimate rose from 21.3% to 58.9% (Table S2; Supplementary Dataset, Instrument_weights). Omitting this variant left non-significant *HLA-G* estimates under either frequency source (Supplementary Dataset, LOO_frequency).

**Colocalization**

*TSHR* PP.H4 (probability of one shared causal variant) was 0.951 in BBJ, 0.986 in FinnGen ophthalmopathy and 0.891 in strict ophthalmopathy, but 0.226 in UKB and 0.264 in FinnGen GD, where PP.H3 (distinct variants) was highest (Table 2; Figures 2 and 3). The UKB and FinnGen GD values were unchanged in narrower regions and in a variant set common to all five datasets, and the matched variants included the strongest regional disease variant in each dataset (Table S3). Lowering the prior probability of a shared variant (p12) from $10^{-5}$ to $10^{-6}$ gave PP.H4 of 0.875 in FinnGen ophthalmopathy, 0.661 in BBJ and 0.451 in strict ophthalmopathy.

*IGF1R* PP.H4 ranged from 0.010 to 0.404 (UKB); in the other four datasets, PP.H2 (expression signal only) was highest (0.571–0.731). *CTLA4* PP.H4 was ≥ 0.80 in UKB and both ophthalmopathy endpoints but not in BBJ or FinnGen GD. None of the 24 genes reached PP.H4 ≥ 0.80 in FinnGen GD, and *HLA-DQA2* and *TUBB* had PP.H4 < 0.001 in every dataset. Apart from *TSHR* and *CTLA4*, only *TNFSF14* and *IFNGR1* reached PP.H4 ≥ 0.80 in any dataset, both in BBJ only (Figure 3).

### Tissue-specific extension

Four of the ten gene–tissue combinations had a qualifying instrument, each a single SNP: *TSHR* in subcutaneous and visceral adipose and *IGF1R* in thyroid and cultured fibroblasts. Thus, 20 of the 50 planned comparisons were estimable and 30 were not (Table S4). All ten adipose *TSHR* ORs were below 1, and eight, including both primary FinnGen GD comparisons, passed the multiple-testing threshold of their test family. The adipose instruments, rs12101255 (subcutaneous) and rs74064796 (visceral), lie 50 bp apart in *TSHR* intron 1, about 15 kb from rs179252 (European $r^2$ 0.46 for rs12101255), and both are blood *TSHR* eQTLs (eQTLGen $P = 4.1\times10^{-34}$ and $6.5\times10^{-34}$). With one instrument, each tissue *P* value equals that variant's disease-association z-test *P* value, so the adipose estimates re-express the regional *TSHR* association. No thyroid or cultured-fibroblast *IGF1R* estimate reached $P < 0.05$, and directions were inconsistent across datasets.

### Descriptive orbital expression

Mean orbital *TSHR*, *IGF1R* and *CTLA4* transcript levels in four patients with inactive TED exceeded those in the single control, whose sample came from a different fat depot; no significance testing was performed (Figure S1).

## Discussion

Higher genetically proxied blood *TSHR* expression was associated with lower GD odds in BBJ and FinnGen GD, with concordant directions for UKB hyperthyroidism and both ophthalmopathy endpoints. All five *IGF1R* ORs exceeded 1, but only BBJ and UKB were nominally significant, none passed the screen threshold, and FinnGen GD, the most precise, lay close to the null. Under the primary prior, colocalization favored a shared *TSHR* variant except in UKB and FinnGen GD. Significance for one gene but not the other is not a formal test of different effects.

Ji and colleagues linked blood *TSHR* expression to GD and strict ophthalmopathy in an earlier FinnGen release [9], and integrative work also investigated *TSHR* [11]. We add an *IGF1R* comparison, GD in two populations and a druggable-gene screen. Both studies used eQTLGen and FinnGen, so agreement is not fully independent. Ji and colleagues accepted PP.H4 above 0.50 [9], whereas we used PP.H4 ≥ 0.80 with p12 = $10^{-5}$; with different methods and releases, the labels are not directly comparable and differences are not contradictions.

The multi-instrument FinnGen and UKB *CTLA4* estimates relied heavily on rs13030124; with that caveat, its direction-consistent associations across all five datasets support it as an autoimmune comparator [26]. *HLA-A*, the strongest BBJ association, was non-significant in FinnGen GD, so the smallest discovery *P* value need not transfer best. *HLA-DQA2* was discordant only in its one two-instrument estimate, UKB, whose instruments were strongly heterogeneous (Supplementary Dataset, MR_estimators_original). With shared instruments, MHC linkage, single-variant dependence of the FinnGen GD *TUBB* and *C4A* estimates, one regional association behind the three chromosome 16p11.2 genes, and no shared-variant support for *HLA-DQA2* or *TUBB*, these associations cannot each be counted as an independent causal target.

The lower-odds direction need not contradict stimulating-antibody pathogenesis: blood transcript abundance, tissue expression and receptor activation are different exposures with tissue-dependent regulatory effects [25]. The expression-increasing rs179252-G allele is linked to rs179247-G, the GD-protective allele associated with greater intrathymic *TSHR* transcription [27, 28]; this fits central tolerance, though the thymic step was untested and neither variant is a proven causal regulator. Each rs179252-G allele raises blood expression by 0.105 standard deviation, and its per-allele disease ORs were 0.80 (BBJ), 0.77 (UKB), 0.78 (FinnGen ophthalmopathy), 0.77 (FinnGen GD) and 0.82 (strict ophthalmopathy);

rescaled to one standard deviation, these moderate effects give the reported odds ratios, which describe lifelong inherited differences, not raised *TSHR* expression or TSHR-directed treatment.

Very small *TSHR* MR *P* values in UKB and FinnGen GD coexisted with distinct-variant patterns, as for *CTLA4* in BBJ and FinnGen GD, so association strength and shared-variant evidence are not interchangeable: a non-zero MR estimate can arise from distinct but correlated variants [29]. These two datasets had the most cases, and their strongest regional disease variant was more strongly associated than rs179252 (Table S3); regional checks excluded a missing-peak explanation but did not identify the causal variant. Support in BBJ and strict ophthalmopathy fell below 0.80 only at the lowest prior tested (Figure 3B). The adipose instruments are themselves blood *TSHR* eQTLs, so their estimates re-express the regional association and cannot resolve the pattern or establish an adipose mechanism. These data establish association, not expression-mediated causality.

In mice, the shed TSHR A-subunit induced more thyroid-stimulating autoimmunity than a minimally cleaving receptor [30]; plasmid immunization reproduced orbital adipogenesis, fibrosis and muscle pathology [31], and orbital fibroblasts from immunized mice showed enhanced inflammatory and hyaluronan responses to TNFα [32]. Ophthalmopathy also occurs in euthyroid and hypothyroid patients [33]. Our instruments measure neither receptor cleavage nor antibody activity. Because most patients with TED have GD, the controls included people with GD but no recorded eye disease, and the ophthalmopathy per-allele ORs resembled those for GD (Table S3), these estimates may mostly reflect GD susceptibility; their higher PP.H4 values, from the two smallest, weakest-signal datasets, do not establish a TED-specific effect.

For *IGF1R*, blood expression-related evidence was limited: colocalization mostly favored an expression-only signal, consistent with weak regional disease signals, and some single-variant omissions removed nominal significance, though most left ORs close to the full estimates (Figure S2). The near-null FinnGen GD estimate argues against large, but not modest, effects; non-significance is not evidence of no inherited role. Single-instrument tissue estimates and absent orbital eQTL data leave tissue effects open. Weaker genetic evidence for *IGF1R* does not argue against IGF-1R blockade: teprotumumab reduced proptosis in a randomized trial in active TED [3] and in long-duration, low-activity TED [34], and TSHR–IGF-1R crosstalk [5] concerns receptor signaling, not inherited susceptibility.

These results inform research rather than treatment selection, leaving unresolved the tissue mediating the *TSHR* association and any inherited *IGF1R* contribution, which independent instruments, thymic and orbital regulatory data and multi-signal colocalization could address. Orbital susceptibility requires comparing GD patients with and without TED, accounting for follow-up and selection into GD.

**Limitations**

BBJ analyses assumed that predominantly European eQTLGen effects and LD transfer to East Asians, in both the MR and colocalization. UKB self-reported hyperthyroidism is broader than GD and was not established as exclusively European. The three overlapping, population-controlled FinnGen endpoints are not independent replications, do not compare GD patients with and without TED, and cannot separate cohort from phenotype differences; participant overlap with eQTLGen was unquantified. Blood is neither thymus nor orbit: the predominantly European GTEx extension had four single-instrument gene–tissue combinations, reused the disease data and lacked colocalization, and adipose and cultured fibroblasts are not orbital tissue, with 30 unestimable comparisons that are not negative

findings. The orbital series (inactive TED; one control from a different fat depot) is descriptive only.

The FinnGen extensions, candidate expansion and colocalization emphasis were post hoc, and only the screen, focused and tissue-family thresholds address specified test families. The single *TSHR* instrument precludes pleiotropy and omission tests and cannot exclude neighboring or correlated variants; other genes have few or heterogeneous instruments, MR-Egger has little power with three or four *IGF1R* instruments, and frequency substitution and omission reused the same data. Under the single-causal-variant model, a low PP.H4 can reflect model mismatch or complex MHC LD; multi-signal colocalization [35], directionality and formal between-phenotype or between-gene tests were not performed, and only 24 genes were assessed. At the screen threshold, most BBJ genes lacked 80% power for an OR of 1.5, so a non-significant MR result, a PP.H4 below 0.80 or an unestimable comparison does not establish absence of an effect.

**Conclusions**

Higher genetically proxied blood *TSHR* expression was associated with lower odds of GD, hyperthyroidism and ophthalmopathy in every dataset. *IGF1R* odds ratios exceeded 1 in every dataset but nominally significant only in BBJ and UKB, and the most precise estimate, FinnGen GD, lay close to the null. Colocalization supported a shared *TSHR* variant in FinnGen ophthalmopathy and, except at the lowest prior tested, in BBJ and strict ophthalmopathy, but favored distinct variants in UKB and FinnGen GD. These analyses addressed disease susceptibility, not treatment response, and the *IGF1R* evidence does not argue against IGF-1R blockade. The ophthalmopathy analyses used population controls and did not estimate TED susceptibility among patients with GD.

## Declarations

**Funding.** This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

**Conflict of interest.** The authors declare that they have no conflict of interest.

**Ethics approval.** This study used publicly available, de-identified summary statistics and an institutionally approved orbital tissue dataset. The in-house orbital transcriptomic component was approved by the Institutional Review Board of Pusan National University Hospital (approval number 2104-018-102) and was conducted in accordance with the Declaration of Helsinki.

**Informed consent.** Written informed consent was obtained from all individual participants included in the in-house orbital tissue study.

**Data availability.** Instruments, primary MR and colocalization results, and sensitivity results are supplied as Supplementary Dataset. Analysis code is available from the corresponding author on reasonable request. Public summary statistics analyzed in this study are available from their original repositories: blood *cis*-eQTL data from the eQTLGen Consortium, tissue eQTL data from GTEx v8, the Biobank Japan Graves disease and UK Biobank hyperthyroidism genome-wide association statistics through the GWAS Catalog, and the FinnGen Release 12 GD and ophthalmopathy statistics from FinnGen. The in-house orbital RNA-seq data are available from the corresponding author on reasonable request, subject to institutional and ethical restrictions.

**Author contributions.** J.P. and M.-S.K. contributed equally to this work. J.P. conceived and designed the study, performed the analyses, and drafted the manuscript. M.-S.K. contributed to data collection and interpretation. S.-W.Y. supervised the study and revised the manuscript.

**Acknowledgements.** We thank the eQTLGen Consortium, the GTEx Consortium, Biobank Japan, the UK Biobank, and the FinnGen study and its participants for making their summary statistics publicly available. We acknowledge the GWAS Catalog for hosting and distributing the genome-wide association summary statistics used as outcome data.

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## Figure legends

**Figure 1. Study design and gene-level discovery in BBJ.** (A) Analysis flow. Blood expression instruments from eQTLGen were tested against BBJ GD for discovery; UKB hyperthyroidism and FinnGen GO were the original comparison datasets, and FinnGen GD, FinnGen GO strict and the GTEx tissue extension were added post hoc. The three FinnGen endpoints overlap and are not independent replications. (B) Each grey point is one estimable gene (n = 2,234), placed at the GRCh37 position of its strongest selected expression instrument and plotted by its BBJ MR *P* value; points are genes, not single variants. Filled black points are the 13 genes that passed $P < 0.05/2{,}544$ (dashed line, $-\log_{10}P \approx 4.71$); the open circle is the focal gene *IGF1R*, which did not. The threshold applies to the BBJ discovery screen, not to later exploratory analyses. GTEx, Genotype-Tissue Expression; MR, Mendelian randomization; SNP, single-nucleotide polymorphism; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

**Figure 2. *TSHR*, *IGF1R* and *CTLA4* across five disease datasets.** (A) MR odds ratios with 95% CIs per approximately 1 SD higher genetically predicted blood expression: Wald ratio for one instrument, multiplicative random-effects IVW for several. The axis is logarithmic and the dashed line is OR = 1. *TSHR* used one instrument, rs179252, in every dataset. Colours identify datasets. (B) Colocalization posterior probabilities for the same comparisons ($p1 = p2 = 10^{-4}$, $p12 = 10^{-5}$), stacked from zero as H4 (one shared variant), H3 (distinct variants), H2 (expression signal only) and H0/H1 (neither trait, or disease only). The dotted line marks PP.H4 = 0.80, the descriptive reporting threshold. Estimates, *P* values and posterior probabilities are given in Table 2; *P* values and posterior probabilities answer different questions. FinnGen GO controls include people with GD but no recorded eye disease. CI, confidence interval; IVW, inverse-variance weighted; MR, Mendelian

randomization; OR, odds ratio; PP, posterior probability; SD, standard deviation; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

**Figure 3. Colocalization across 24 selected genes and *TSHR* prior sensitivity.** (A) PP.H4 for each gene and dataset ($p1 = p2 = 10^{-4}$, $p12 = 10^{-5}$); shading follows the colour bar. Boxed cells have PP.H4 ≥ 0.80, and a dot marks cells that remain ≥ 0.80 at $p12 = 10^{-6}$. Horizontal lines separate *TSHR*, *IGF1R* and *CTLA4*; the other 11 BBJ discoveries; the nine additional FinnGen GD discoveries; and *HLA-G*, added because its MR result changed with the allele-frequency source. Only these 24 genes underwent colocalization. Values below 0.001 are shown as <0.001; full-precision values are in the Supplementary Dataset. Low BBJ PP.H4 for the two *IGLV* genes partly reflects absence of the lead expression variant from the BBJ data. (B) *TSHR* PP.H4 in each dataset at three shared-variant priors (p12); points are offset slightly to avoid overlap, and the dashed line marks 0.80. PP.H4 is a posterior probability under the colocalization model, not a *P* value. MR, Mendelian randomization; PP.H4, posterior probability of one shared causal variant; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table 1. Data sources and their roles

| Dataset / endpoint | Cases | Controls | Role |
|---|---|---|---|
| eQTLGen blood cis-eQTL | — | — | Exposure; up to 31,684 participants (median 28,092 per variant); predominantly European |
| BBJ GD (GCST90018627) | 2,809 | 172,656 | Discovery; East Asian |
| FinnGen GD (E4_GRAVES_STRICT) | 3,962 | 496,386 | Post hoc same-disease comparison; Finnish |
| UKB hyperthyroidism (GCST90038636) | 3,731 | 480,867 | Broader-phenotype comparison; self-reported hyperthyroidism or thyrotoxicosis; ancestry not established as exclusively European |
| FinnGen GO (GRAVES_OPHT) | 858 | 499,490 | Original ophthalmopathy (TED-enriched) comparison; Finnish |
| FinnGen GO strict (E4_GRAVES_OPHT_STRICT) | 753 | 499,595 | Post hoc ophthalmopathy definition sensitivity; Finnish |
| GTEx v8 tissue eQTL | — | — | Post hoc tissue extension (thyroid, cultured fibroblasts, subcutaneous and visceral adipose, whole blood; n = 469–670; Table S4) |
| 1000 Genomes Phase 3 European reference (n = 503) | — | — | LD clumping of instruments and reference allele frequencies |

The three FinnGen R12 endpoints share participants and are not independent replications. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. UKB hyperthyroidism is broader than GD. ‘Strict’ is the FinnGen endpoint label, not proof of perfect specificity. The descriptive orbital RNA-sequencing series (four TED patients, one control) is separate from these datasets. eQTL, expression quantitative trait locus; GTEx, Genotype-Tissue Expression; LD, linkage disequilibrium; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table 2. MR and colocalization results for *TSHR*, *IGF1R* and *CTLA4* across five disease datasets

| Gene | Dataset | IVs | OR (95% CI) | *P* value | Q-test *P* | PP.H2 | PP.H3 | PP.H4 | PP.H4, p12 = $10^{-6}$ |
|---|---|---|---|---|---|---|---|---|---|
| *TSHR* | BBJ GD | 1 | 0.12 (0.07–0.21) | $1.09\times10^{-14}$*† | NA | <0.001 | 0.049 | **0.951** | 0.661 |
| *TSHR* | FinnGen GD | 1 | 0.08 (0.05–0.12) | $3.19\times10^{-32}$*† | NA | <0.001 | 0.736 | 0.264 | 0.035 |
| *TSHR* | UKB hyperthyroidism | 1 | 0.09 (0.06–0.14) | $8.77\times10^{-28}$*† | NA | <0.001 | 0.774 | 0.226 | 0.028 |
| *TSHR* | FinnGen GO | 1 | 0.10 (0.04–0.24) | $2.82\times10^{-7}$*† | NA | <0.001 | 0.014 | **0.986** | **0.875** |
| *TSHR* | FinnGen GO strict | 1 | 0.15 (0.06–0.39) | $9.02\times10^{-5}$* | NA | 0.027 | 0.082 | **0.891** | 0.451 |
| *IGF1R* | BBJ GD | 4 | 1.56 (1.07–2.28) | 0.0212* | 0.902 | 0.690 | 0.236 | 0.073 | 0.008 |
| *IGF1R* | FinnGen GD | 3 | 1.07 (0.86–1.33) | 0.539 | 0.882 | 0.731 | 0.259 | 0.010 | 0.001 |
| *IGF1R* | UKB hyperthyroidism | 4 | 1.35 (1.07–1.70) | 0.0117* | 0.250 | 0.400 | 0.196 | 0.404 | 0.063 |
| *IGF1R* | FinnGen GO | 3 | 1.41 (0.85–2.33) | 0.182 | 0.309 | 0.623 | 0.346 | 0.032 | 0.003 |
| *IGF1R* | FinnGen GO strict | 3 | 1.52 (0.92–2.49) | 0.0996 | 0.409 | 0.571 | 0.391 | 0.038 | 0.004 |
| *CTLA4* | BBJ GD | 1 | 0.18 (0.11–0.27) | $5.45\times10^{-15}$*† | NA | <0.001 | 0.799 | 0.201 | 0.025 |
| *CTLA4* | FinnGen GD | 2 | 0.17 (0.08–0.36) | $2.15\times10^{-6}$*† | 0.0275 | <0.001 | 0.746 | 0.254 | 0.033 |
| *CTLA4* | UKB hyperthyroidism | 2 | 0.21 (0.08–0.57) | 0.00220* | 0.00131 | <0.001 | 0.047 | **0.953** | 0.672 |
| *CTLA4* | FinnGen GO | 2 | 0.17 (0.04–0.68) | 0.0118* | 0.0486 | <0.001 | 0.022 | **0.978** | **0.816** |
| *CTLA4* | FinnGen GO strict | 2 | 0.17 (0.05–0.64) | 0.00888* | 0.0755 | <0.001 | 0.021 | **0.979** | **0.820** |

MR estimates are ORs per approximately one SD higher genetically predicted blood expression: Wald ratio for one IV, multiplicative random-effects IVW for several. *TSHR* used one instrument, rs179252, in every dataset; its G allele raises blood expression by 0.105 SD per allele (F = 176), and its per-allele disease ORs were 0.77–0.82 (Table S3). The per-SD ORs therefore rescale a moderate per-allele association, and each single-IV *P* value equals the variant's disease-association z-test *P* value. *CTLA4* multi-IV estimates are concentrated in rs13030124, and three of the four have Q-test $P < 0.05$; in BBJ and UKB, two of four single-instrument omissions removed nominal *IGF1R* significance (Figure S2). * Nominal $P < 0.05$; † $P < 0.05/2{,}544$, the screen threshold, applied descriptively outside BBJ. PP.H2–H4 are colocalization posterior probabilities (H2, expression signal only; H3, distinct variants; H4, one shared variant) with $p1 = p2 = 10^{-4}$ and $p12 = 10^{-5}$ unless stated; bold, ≥0.80. They are not *P* values. BBJ colocalization pairs European eQTL with East Asian disease statistics. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. CI, confidence interval; eQTL, expression quantitative trait locus; IV, instrumental variable; IVW, inverse-variance weighted; MR, Mendelian randomization; NA, not applicable (one IV); OR, odds ratio; PP, posterior probability; SD, standard deviation; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table 3. Thirteen BBJ discoveries and their FinnGen GD comparison

| BBJ discovery gene | Chr | IVs, BBJ/FinnGen GD | BBJ OR (95% CI) | BBJ *P* | FinnGen GD OR (95% CI) | FinnGen GD *P* | Focused support | FinnGen GD PP.H4 |
|---|---|---|---|---|---|---|---|---|
| *HLA-A* | 6 (MHC) | 4/4 | 1.98 (1.73–2.27) | $2.60\times10^{-23}$ | 1.08 (0.77–1.53) | 0.648 | No | <0.001 |
| *HLA-DQA2* | 6 (MHC) | 1/1 | 2.39 (1.95–2.93) | $8.30\times10^{-17}$ | 1.90 (1.64–2.20) | $1.41\times10^{-17}$ | Yes | <0.001 |
| *CTLA4* | 2 | 1/2 | 0.18 (0.11–0.27) | $5.45\times10^{-15}$ | 0.17 (0.08–0.36) | $2.15\times10^{-6}$ | Yes | 0.254 |
| *TSHR* | 14 | 1/1 | 0.12 (0.07–0.21) | $1.09\times10^{-14}$ | 0.08 (0.05–0.12) | $3.19\times10^{-32}$ | Yes | 0.264 |
| *C4A* | 6 (MHC) | 1/3 | 0.45 (0.35–0.57) | $2.23\times10^{-10}$ | 0.41 (0.26–0.63) | $4.97\times10^{-5}$ | Yes | <0.001 |
| *HSD3B7* | 16 (16p11.2 cluster) | 1/2 | 0.29 (0.18–0.47) | $2.04\times10^{-7}$ | 0.90 (0.72–1.13) | 0.372 | No | 0.011 |
| *TUBB* | 6 (MHC) | 2/3 | 1.49 (1.28–1.74) | $3.14\times10^{-7}$ | 3.28 (2.09–5.17) | $2.68\times10^{-7}$ | Yes | <0.001 |
| *VKORC1* | 16 (16p11.2 cluster) | 1/0 | 0.13 (0.06–0.29) | $6.35\times10^{-7}$ | NA | NA | NA | 0.011 |
| *TNFSF14* | 19 | 1/1 | 0.63 (0.52–0.76) | $1.48\times10^{-6}$ | 0.77 (0.65–0.92) | 0.00361 | Yes | 0.303 |
| *PRSS36* | 16 (16p11.2 cluster) | 1/1 | 6.00 (2.87–12.57) | $2.02\times10^{-6}$ | 1.40 (1.00–1.96) | 0.0490 | No | 0.063 |
| *MAPKAPK5* | 12 | 4/4 | 0.977 (0.967–0.987) | $5.28\times10^{-6}$ | 0.990 (0.929–1.055) | 0.751 | No | 0.041 |
| *PSMB8* | 6 (MHC) | 1/1 | 1.25 (1.13–1.38) | $6.86\times10^{-6}$ | 0.96 (0.85–1.08) | 0.512 | No | <0.001 |
| *IFNGR1* | 6 | 1/1 | 2.10 (1.51–2.91) | $9.41\times10^{-6}$ | 1.62 (1.18–2.24) | 0.00330 | Yes | 0.313 |

Genes passed the BBJ screen threshold ($P < 0.05/2{,}544$) and are listed by BBJ *P* value. Focused support in FinnGen GD requires the same direction and $P < 0.05/13$; 7 of 13 qualify. IVs are counted separately for each dataset because instrument availability differs. The five MHC genes and the three chromosome 16p11.2 genes are not independent signals; the latter are compatible with one regional association. *MAPKAPK5* is shown to three decimals to preserve its small effect. NA, not estimable (no retained instrument), not a negative result; *VKORC1* colocalization does not require an MR instrument. PP.H4, shared-variant posterior probability under $p12 = 10^{-5}$; bold, ≥0.80. Chr, chromosome of the strongest instrument; MHC, major histocompatibility complex; CI, confidence interval; IV, instrumental variable; MR, Mendelian randomization; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Figure 1

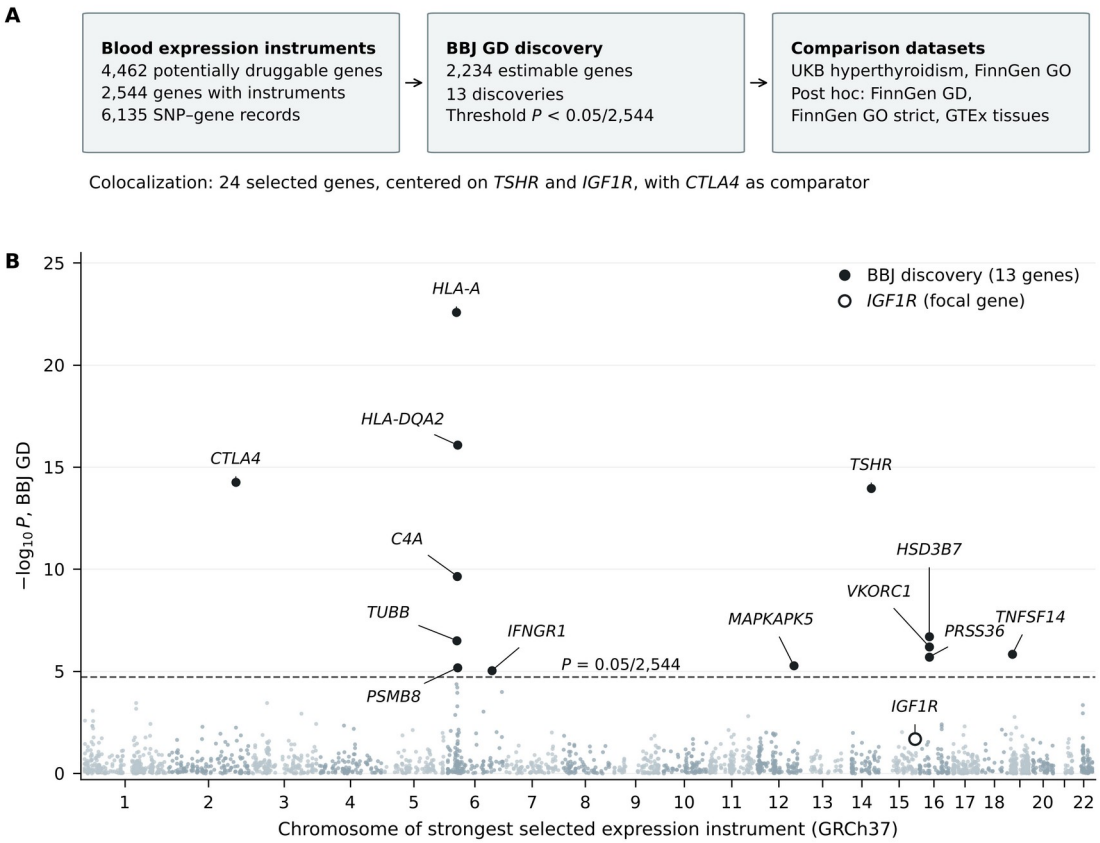



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## Figure 2

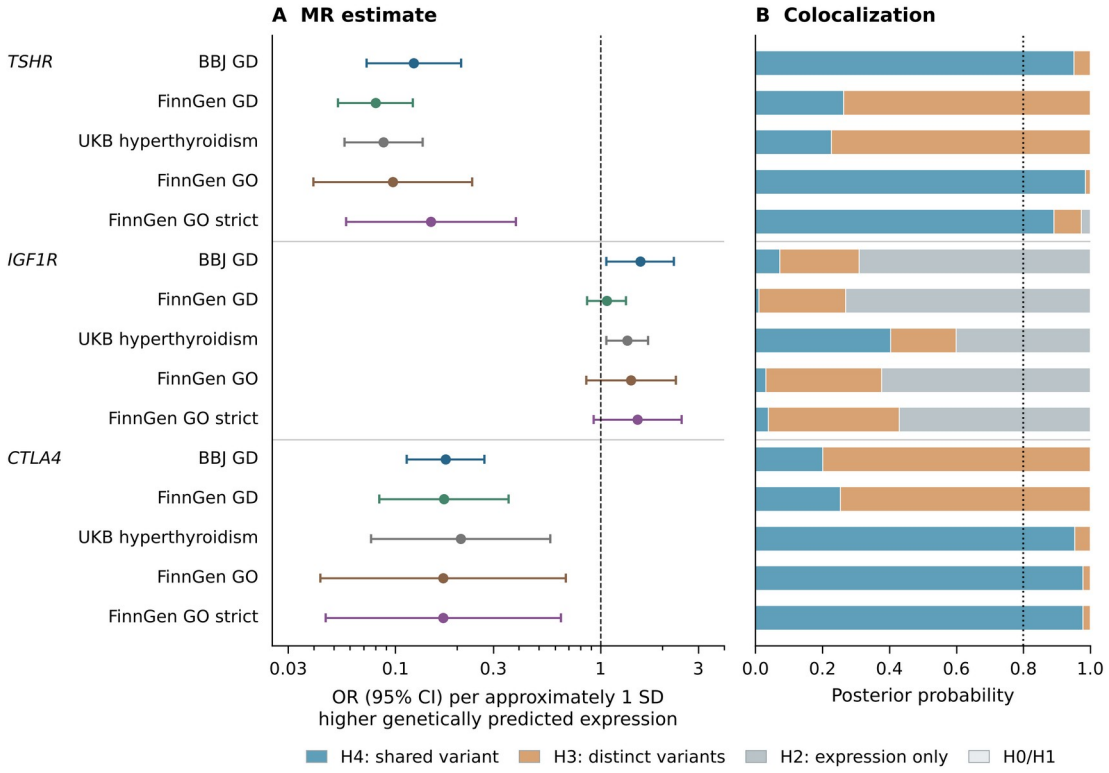



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## Figure 3

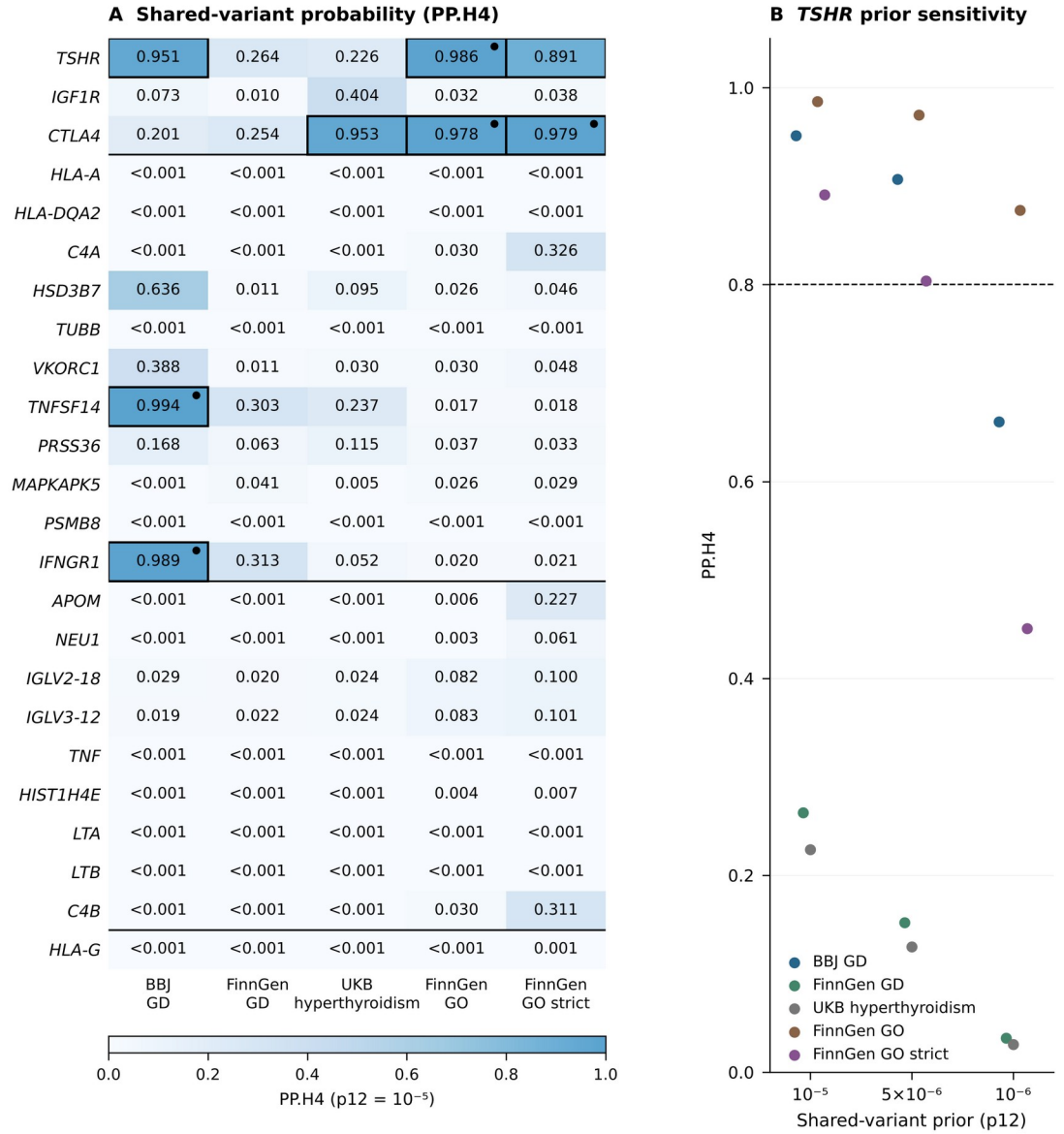



**Figure 3. Colocalization across 24 selected genes and *TSHR* prior sensitivity.** (A) PP.H4 for each gene and dataset (p1 = p2 = $10^{-4}$, p12 = $10^{-5}$); shading follows the colour bar. Boxed cells have PP.H4 ≥ 0.80, and a dot marks cells that remain ≥ 0.80 at p12 = $10^{-6}$. Horizontal lines separate *TSHR*, *IGF1R* and *CTLA4*; the other 11 BBJ discoveries; the nine additional FinnGen GD discoveries; and *HLA-G*, added because its MR result changed with the allele-frequency source. Only these 24 genes underwent colocalization. Values below 0.001 are shown as <0.001; full-precision values are in the Supplementary Dataset. Low BBJ PP.H4 for the two *IGLV* genes partly reflects absence of the lead expression variant from the BBJ data. (B) *TSHR* PP.H4 in each dataset at three shared-variant priors (p12); points are offset slightly to avoid overlap, and the dashed line marks 0.80. PP.H4 is a posterior probability under the colocalization model, not a *P* value. MR, Mendelian randomization; PP.H4, posterior probability of one shared causal variant; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Supplementary material

TSHR and IGF1R in Graves disease and thyroid eye disease: genetic evidence from blood expression

## Supplementary methods

### Dataset definitions and analysis chronology

BBJ GD (GCST90018627) and UKB self-reported hyperthyroidism/thyrotoxicosis (GCST90038636) retained the source sample sizes and estimates. UKB used BOLT-LMM linear-model statistics; its report included adjustment for sex, age, BMI, assessment centre, ethnicity, genotyping batch and 20 principal components. The available documentation did not establish an exclusively European analysis sample. FinnGen R12 GWAS counts were 858/499,490 for GRAVES_OPHT, 3,962/496,386 for E4_GRAVES_STRICT and 753/499,595 for E4_GRAVES_OPHT_STRICT. Counts refer to the analyzed GWAS files, not later phenotype-browser totals. Strict GD combines specified GD/ophthalmopathy diagnostic criteria with the Graves condition and exclusion of amiodarone exposure; strict ophthalmopathy requires its specified ophthalmopathy and thyrotoxicosis codes. BBJ cases merged physician diagnoses with curated medical histories and text-mined records; controls lacked GD or related diagnoses; covariates were age, age$^2$, sex, their interactions and 20 principal components. None supplies GD-only controls.

The original three-outcome analysis preceded the two FinnGen extensions. The expanded analysis plan was recorded before the new endpoint associations were computed, but it was not a prospective registration of the complete study. The 24 colocalization genes comprise 13 BBJ discoveries, *IGF1R*, nine additional FinnGen GD screen discoveries, and frequency-sensitive *HLA-G*. No new discovery threshold was optimized from these results.

### Instrument selection and effect reconstruction

Of 4,479 druggable-genome genes, 4,462 mapped to GRCh37. Significant cis-eQTL instruments ($P < 5\times10^{-8}$ within ±1 Mb of the gene) were clumped at $r^2 < 0.001$ within 10 Mb using the European 1000 Genomes Phase 3 reference (n = 503). $SE = 1/\sqrt{[2f(1-f)(N + Z^2)]}$ and $\beta = Z \times SE$, where f is the assessed-allele frequency and N is the variant-specific expression sample size (median 28,092; range 232–31,684). Unit expression variance was assumed. Variant $F = Z^2$. The estimate is approximately per expression SD, not per SD of the genetic predictor.

TwoSampleMR harmonise_data(action = 2) aligned alleles and excluded ambiguous palindromic variants with MAF > 0.42. Missing outcome estimates were not imputed. BBJ and FinnGen retained log-odds scales; UKB β and SE were each divided once by $\mu(1-\mu)$, with $\mu = 3{,}731/484{,}598$. GRCh37 expression and GRCh38 FinnGen coordinates were not directly joined: identifiers and allele pairs

were checked, with duplicate and alias audits. Source covariate adjustments were retained; patient-level re-adjustment and subgroup analysis were unavailable.

**MR estimators and instrument diagnostics**

For a single instrument, the Wald coefficient is $\beta_Y/\beta_X$ and its first-order SE is $SE_Y/|\beta_X|$. Multiple-instrument MR used inverse-variance weighted regression through the origin. Multiplicative random-effects SEs included under-dispersion correction, so heterogeneity could widen the interval. Normal two-sided tests and 95% CIs were used for the primary estimators. The alternative-estimator worksheet preserves weighted median, weighted mode and MR-Egger results for the original three outcomes; mode and Egger intervals use the corresponding t distributions. These alternative estimators were not rerun for all new endpoints. With three or four *IGF1R* instruments, these estimators had little power. No Steiger directionality test was performed.

Post hoc leave-one-out analyses applied IVW when at least two variants remained and the Wald ratio when one remained. Single-instrument *TSHR* and BBJ *CTLA4* were ineligible. Original three-outcome results, new-endpoint results for the fixed 14-gene set, and 72 *C4B*/*HLA-G* omission estimates under two frequency sources are retained in separately named worksheets. Independent calculations reproduced the eight full-set and 72 omission estimates for *C4B* and *HLA-G*. They reuse the same association data and are not independent replications. *C4B*/*HLA-G* instrument weights use $\beta_X^2/SE_Y^2$, normalized within each full model. A single-SNP *P* value can be much smaller than the full IVW *P* value when heterogeneity inflates the full-model SE.

*TSHR* clumping sensitivity used eQTL thresholds of $5\times10^{-8}/5\times10^{-6}$, $r^2$ thresholds of 0.001/0.01/0.1, and European (n = 503) or East Asian (n = 504) reference panels. Primary settings selected one European and two East Asian variants, and rs179252 was retained in every setting. Chromosome 16p11.2 analyses found high East Asian LD among rs4889606, rs34649473 and rs78924645; GCTA 1.95.1 COJO stepwise selection (BBJ; East Asian reference) retained one signal, rs8050588; conditional on it, the three variants had *P* = 0.81–0.93, supporting regional dependence rather than three independent gene effects.

**TSHR allele alignment and reference LD**

In a post hoc audit, original full eQTLGen and disease GWAS files were re-extracted for rs179252 and rs179247. Their GRCh37 positions were 14:81435985 and 14:81432546, and GRCh38 positions were 14:80969641 and 14:80966202. Both non-palindromic allele pairs matched across sources. UKB effects were reversed to the G allele after the existing scale conversion; BBJ and FinnGen required no reversal. Exposure coefficients and first-order Wald estimates were independently reconstructed. Original GWAS *P* values and recalculated MR *P* values were retained separately.

G-allele counts at both variants were positively correlated in European (n = 503; r = 0.978322812; $r^2$ = 0.957115524) and East Asian (n = 504; r = 0.995803919; $r^2$ = 0.991625445) references, without missing genotypes. Direct BED decoding agreed with PLINK dosage output. PLINK haplotype estimates supported G–G and T–A coupling; these were inferred from unphased genotypes, not directly observed phased haplotypes. Alignment to the G allele resolved the negative default European correlation obtained when counting T at rs179252. The TSHR_allele_audit worksheet retains both eQTL records, both variants across five disease outcomes, all four inferred haplotypes in each population, and MR reproduction. This audit does not identify a causal variant or establish mediation in thymus or orbit.

**Colocalization hypotheses and regional inputs**

coloc.abf received beta/varbeta data for disease (dataset 1, case-control type) and expression (dataset 2, quantitative type, sdY = 1), with the default prior effect SDs of 0.2 and 0.15, respectively. Outcome allele frequency was not used in the Bayes factors. Complete available cis-eQTL statistics, not just significant eQTLs, were matched by identifier and allele compatibility. H0 denotes neither trait associated; H1 disease only; H2 expression only; H3 both traits associated through distinct variants; H4 both traits associated through one shared variant. PP.Hk is the posterior probability of hypothesis k, conditional on the model and priors.

Primary priors were p1 = p2 = $10^{-4}$ and p12 = $10^{-5}$; sensitivity p12 values were $5\times10^{-6}$ and $10^{-6}$. The PP.H4 ≥ 0.80 threshold was a reporting convention, not a significance test. H3 is a competing distinct-variant model, not a count of causal signals. Multi-signal colocalization was not performed.

Four frequency/availability scenarios were used: original_reference (original available reference frequencies), paired_reference and paired_eqtlgen (the same variants with the respective frequencies), and available_eqtlgen (all matching variants with valid cohort frequencies). There are 1,440 gene–outcome–scenario–prior rows; no PP.H4 changed side of 0.80 across scenarios. The additional *TSHR* analysis comprises four windows × five outcomes × four scenarios × three priors = 240 rows. Windows were the full source cis-region, ±500 kb and ±250 kb around the recorded GRCh37 GenePos anchor, and the five-outcome variant intersection. The full cis-source contained 7,309 eQTL rows; all three FinnGen full-region analyses matched 6,579 variants, and the five-outcome intersection contained 4,030. Raw outcome regional scans checked whether the strongest disease variants survived matching. New candidate coverage checks also examined marginal expression strength and shared instruments; the absent rs5759410 lead in BBJ was checked against variant aliases without finding a replacement.

For *TSHR*, *IGF1R* and *CTLA4*, colocalization and MR in FinnGen GD and strict ophthalmopathy were independently reproduced from re-downloaded FinnGen R12 regional statistics using two

separately written implementations, with the original ophthalmopathy endpoint as a control. The allele-compatible variant sets were the same (6,675, 7,850 and 4,993 variants), and the top variant was the same in all 27 gene–endpoint–prior settings, including the control. The reproduction used gnomAD v2.1.1 European frequencies, which were available for more of these variants than the 1000 Genomes European frequencies used in the reported analysis (6,674, 7,850 and 4,992 analysed, versus 6,579, 7,740 and 4,924). On the control endpoint, PP.H4 changed by at most $4.1\times10^{-4}$; for *TSHR* this difference came from the additional variants. Reported new-endpoint PP.H4 values were reproduced within $1.2\times10^{-3}$, and no setting changed side of 0.80. The *TSHR* strict-ophthalmopathy value at p12 = $10^{-6}$ was 0.449 rather than 0.451 (0.451 on the reported variant set), and *IGF1R* PP.H2 and PP.H3 differed by up to 0.003, as on the control endpoint. MR estimates were reproduced to numerical precision on the original exposure scale.

**Frequency substitution and precision benchmarks**

The published eQTLGen frequency release covered 26,609 participants, excluding Framingham. Reconstructing $\beta_X$ and $SE_X$ with fixed Z and N preserves their ratio up to floating-point precision. For a single-IV Wald estimate, frequency scaling changes the effect unit and SE together and preserves the normal-test *P* value; for IVW, variant weights can change. The new-endpoint MR dataset separates identical-variant scaling from repeat allele harmonization. *C4B* and *HLA-G* were audited with identical retained variants and outcome effects, so their differences arise from scaling and weighting rather than a changed variant set. Large frequency discrepancies are documented, not assumed to be resolved by correct identifier matching.

The detection benchmarks use $(z_{1-\alpha/2} + z_{0.80}) \times SE$ as the approximate 80% minimum detectable absolute log-OR. Alpha was 0.05/2,544 for BBJ and 0.05 for the original UKB/FinnGen comparisons. These are precision summaries using observed SEs, not prospective sample-size calculations, post hoc proof of power, or colocalization power. The Detection_original worksheet retains these three-outcome results. Applying the same formula to the MR_primary standard errors for FinnGen GD at alpha 0.05/2,544 gave 24.8% and 52.2% of 2,480 estimable genes with 80% power for ORs of 1.5 and 2.0. At alpha 0.05, the detectable *IGF1R* ORs were 1.72 in BBJ, 1.39 in UKB and 1.36 in FinnGen GD.

**Fixed-panel tissue extension**

The post hoc tissue extension retained a frozen panel of *TSHR* and *IGF1R* in five standard-release GTEx v8 tissues. QTL sample counts were 574 (thyroid), 483 (cultured fibroblasts), 581 (subcutaneous adipose), 469 (visceral adipose) and 670 (whole blood). The original v8 archive supplied significant variant–gene pairs and eGenes records for all tested genes; gene-specific release thresholds exceeded $5\times10^{-8}$ for every eligible combination, permitting complete selection at the fixed

threshold. Absence from the tested-gene file was distinguished from a minimum eQTL *P* value above the threshold. The standard release includes multiple ancestries; the overall donor cohort was 85.3% European American, not a tissue-specific percentage.

Variants were restricted to ±1 Mb of the source GENCODE v26 TSS, $P < 5\times10^{-8}$, F > 10 and MAF ≥ 0.01. GTEx variant IDs identify GRCh38 REF and ALT, with slope referring to ALT. Actual effect-allele frequencies were reconstructed from each selected eGene MAF and ref_factor, not by treating MAF as ALT frequency. GTEx mappings linked GRCh38 variants to rsIDs and GRCh37 reference coordinates. PLINK clumping used 503 European reference samples, $r^2 < 0.001$ and a 10-Mb window; an independent genotype-based calculation reproduced selection. One of 20 thyroid *IGF1R* candidates was absent from the reference, while its strongest candidate was retained. All selected variants were non-palindromic and matched exactly in the five disease datasets; no proxies or relaxed thresholds were used.

Each of the four estimable gene–tissue combinations retained one SNP. Wald estimates used $\beta_Y/\beta_X$ and $SE_Y/|\beta_X|$ with normal two-sided *P* values and 95% CIs; independent calculations reproduced all 20 estimates. The UKB linear-to-log-odds conversion was applied once. ORs represent one unit of inverse-normal-transformed normalized expression; they are not directly comparable to an intervention or to the reconstructed eQTLGen scale. Primary adjusted $P = \min(1, 10P)$ for FinnGen GD; secondary adjusted $P = \min(1, 40P)$ for the other datasets. The 30 unestimable cells remained in the planned families. Single-instrument analyses cannot support heterogeneity, leave-one-out or MR-Egger diagnostics. Significant-only eQTL pairs were not used for colocalization; full regional data were not obtained from the requester-pays resource, and no tissue posterior was calculated. Tissue_extension in the single Supplementary Dataset retains all 50 cells, availability, selected and excluded candidates, harmonized alleles and source statistics.

**Orbital expression and reproducibility**

Orbital adipose tissue was collected during orbital decompression from four patients with inactive TED; the control was nasal fat pad from one person without thyroid disease undergoing cosmetic upper blepharoplasty. Paired-end RNA sequencing (Illumina NovaSeq) used two technical-replicate libraries per donor, which were combined by biological sample; reads were trimmed (Trimmomatic 0.38), aligned to GRCh38 (HISAT2 2.1.0) and quantified as TPM (StringTie 2.1.3b). Figure S1 shows sample-level transcript abundance and $\log_2[(\text{mean TED TPM} + 0.01)/(\text{control TPM} + 0.01)]$. One control cannot estimate control-group biological variance reliably, and no differential-expression *P* value was used. The dataset is descriptive and did not influence candidate selection.

The workbook Readme and Column_dictionary sheets describe every sheet; reproduction validates implementation, not causal assumptions.

## Table S1. Additional FinnGen GD screen discoveries

| Gene | Chr | FinnGen GD IVs | FinnGen GD OR (95% CI) | FinnGen GD *P* | BBJ PP.H4 | FinnGen GD PP.H4 | FinnGen GO strict PP.H4 |
|---|---|---|---|---|---|---|---|
| *APOM* | 6 (MHC) | 1 | 0.05 (0.04–0.06) | $1.43\times10^{-108}$ | <0.001 | <0.001 | 0.227 |
| *NEU1* | 6 (MHC) | 1 | 0.04 (0.03–0.06) | $1.43\times10^{-108}$ | <0.001 | <0.001 | 0.061 |
| *IGLV2-18* | 22 | 1 | 0.56 (0.48–0.66) | $7.52\times10^{-12}$ | 0.029 | 0.020 | 0.100 |
| *IGLV3-12* | 22 | 1 | 1.20 (1.14–1.26) | $7.52\times10^{-12}$ | 0.019 | 0.022 | 0.101 |
| *TNF* | 6 (MHC) | 1 | 0.60 (0.51–0.71) | $4.65\times10^{-10}$ | <0.001 | <0.001 | <0.001 |
| *HIST1H4E* | 6 (MHC) | 1 | 2.54 (1.88–3.43) | $1.51\times10^{-9}$ | <0.001 | <0.001 | 0.007 |
| *LTA* | 6 (MHC) | 1 | 0.46 (0.35–0.61) | $3.75\times10^{-8}$ | <0.001 | <0.001 | <0.001 |
| *LTB* | 6 (MHC) | 1 | 0.18 (0.10–0.36) | $4.89\times10^{-7}$ | <0.001 | <0.001 | <0.001 |
| *C4B* | 6 (MHC) | 2 | 1.63 (1.31–2.03) | $1.10\times10^{-5}$ | <0.001 | <0.001 | 0.311 |

Genes outside the BBJ discovery set that passed $P < 0.05/2{,}544$ in the post hoc full FinnGen GD screen (1000 Genomes European reference frequencies). These are not independent of the screen and seven of nine are in the MHC. *APOM*/*NEU1* share rs3101018 and *IGLV2-18*/*IGLV3-12* share rs5759410. The lead *IGLV* expression variant is absent in BBJ, weakening interpretation of its low PP.H4. None reached PP.H4 ≥0.80 ($p12 = 10^{-5}$) in any dataset. Chr, chromosome; CI, confidence interval; IV, instrumental variable; MHC, major histocompatibility complex; OR, odds ratio; PP.H4, shared-variant posterior probability; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table S2. Frequency-sensitive exploratory candidates

| Gene | Dataset | Frequency source | IVs | OR (95% CI) | *P* value | Q-test *P* |
|---|---|---|---|---|---|---|
| *C4B* | FinnGen GD | 1000G EUR | 2 | 1.63 (1.31–2.03) | $1.10\times10^{-5}$ | $7.79\times10^{-7}$ |
| *C4B* | FinnGen GD | eQTLGen cohort | 2 | 1.75 (1.31–2.33) | $1.48\times10^{-4}$ | $1.36\times10^{-8}$ |
| *C4B* | FinnGen GO strict | 1000G EUR | 2 | 1.68 (1.44–1.95) | $2.74\times10^{-11}$ | 0.122 |
| *C4B* | FinnGen GO strict | eQTLGen cohort | 2 | 1.80 (1.46–2.23) | $6.17\times10^{-8}$ | 0.0587 |
| *HLA-G* | FinnGen GD | 1000G EUR | 16 | 0.86 (0.72–1.03) | 0.107 | $9.98\times10^{-24}$ |
| *HLA-G* | FinnGen GD | eQTLGen cohort | 16 | 0.60 (0.47–0.75) | $1.72\times10^{-5}$ | $2.26\times10^{-10}$ |
| *HLA-G* | FinnGen GO strict | 1000G EUR | 16 | 0.91 (0.76–1.08) | 0.267 | 0.0231 |
| *HLA-G* | FinnGen GO strict | eQTLGen cohort | 16 | 0.70 (0.53–0.92) | 0.0122 | 0.131 |

Exposure effects were reconstructed with 1000 Genomes Phase 3 European reference allele frequencies (1000G EUR; primary) or eQTLGen cohort frequencies; variant sets and outcome effects were held fixed, so changes reflect rescaled expression effects and IVW weights. *HLA-G* was added because its eQTLGen-frequency *P* crossed the screen threshold ($P < 0.05/2{,}544$). Q tests heterogeneity, not directional pleiotropy; its *P* value is not a gene-association *P* value. All 72 omission estimates and instrument weights are in the workbook. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table S3. *TSHR* regional colocalization and prior sensitivity

| Dataset | Matched variants | Disease lead (*P*) | rs179252 *P* | rs179252-G OR (95% CI) | H4 top variant | PP.H4, $10^{-5}$ / $10^{-6}$ | PP.H4 range across windows |
|---|---|---|---|---|---|---|---|
| BBJ GD | 4,336 | rs28414437 ($1.83\times10^{-15}$) | $1.09\times10^{-14}$ | 0.80 (0.76–0.85) | rs179252 | **0.951** / 0.661 | 0.951–0.951 |
| FinnGen GD | 6,579 | rs1023586 ($1.58\times10^{-36}$) | $3.19\times10^{-32}$ | 0.77 (0.73–0.80) | rs1023586 | 0.264 / 0.035 | 0.264–0.264 |
| UKB hyperthyroidism | 5,667 | rs4903961 ($1.17\times10^{-43}$) | $8.77\times10^{-28}$ | 0.77 (0.74–0.81) | rs1023586 | 0.226 / 0.028 | 0.226–0.226 |
| FinnGen GO | 6,579 | rs3783947 ($1.23\times10^{-7}$) | $2.82\times10^{-7}$ | 0.78 (0.71–0.86) | rs179252 | **0.986** / **0.875** | 0.986–0.987 |
| FinnGen GO strict | 6,579 | rs2110695 ($1.34\times10^{-5}$) | $9.02\times10^{-5}$ | 0.82 (0.74–0.90) | rs179252 | **0.891** / 0.451 | 0.891–0.905 |

Disease lead, the strongest disease-associated variant among the matched cis variants, with its *P* value; rs179252 *P*, z-test *P* of the aligned G-allele disease association (the UKB source BOLT-LMM *P* was $1.3\times10^{-27}$). H4 top variant, the variant with the highest posterior under the shared-variant hypothesis; it is not necessarily the disease lead. PP.H4 under p12 = $10^{-5}$ and $10^{-6}$ (p1 = p2 = $10^{-4}$); bold, ≥0.80. Window range covers the full source cis-region, ±500 kb, ±250 kb and the five-outcome intersection at p12 = $10^{-5}$; it is not a confidence interval. All 240 window settings are in TSHR_windows. BBJ pairs European eQTL with East Asian disease statistics. CI, confidence interval; eQTL, expression quantitative trait locus; OR, odds ratio; PP.H4, shared-variant posterior probability; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table S4. *TSHR* and *IGF1R* tissue expression extension

### A. Availability

| Gene / tissue | GTEx N | Instrument (F) | Status |
|---|---|---|---|
| *TSHR* / Thyroid | 574 | — | No instrument (minimum eQTL $P = 9.60\times10^{-8}$) |
| *IGF1R* / Thyroid | 574 | rs1815009 (F = 61.1) | Estimable |
| *TSHR* / Cultured fibroblasts | 483 | — | Not tested by GTEx in this tissue |
| *IGF1R* / Cultured fibroblasts | 483 | rs6598538 (F = 33.8) | Estimable |
| *TSHR* / Subcutaneous adipose | 581 | rs12101255 (F = 38.5) | Estimable |
| *IGF1R* / Subcutaneous adipose | 581 | — | No instrument (minimum eQTL $P = 3.56\times10^{-7}$) |
| *TSHR* / Visceral adipose | 469 | rs74064796 (F = 32.1) | Estimable |
| *IGF1R* / Visceral adipose | 469 | — | No instrument (minimum eQTL $P = 7.76\times10^{-6}$) |
| *TSHR* / Whole blood | 670 | — | No instrument (minimum eQTL $P = 2.46\times10^{-4}$) |
| *IGF1R* / Whole blood | 670 | — | No instrument (minimum eQTL $P = 5.42\times10^{-5}$) |

## B. Estimates

| Gene / tissue | Dataset | Family | OR (95% CI) | *P* value | Below family threshold | Direction vs blood |
|---|---|---|---|---|---|---|
| *TSHR* / Subcutaneous adipose | BBJ GD | Secondary | 0.46 (0.38–0.55) | $1.92\times10^{-15}$ | Yes | Same |
| *TSHR* / Subcutaneous adipose | FinnGen GD | Primary | 0.36 (0.31–0.43) | $1.28\times10^{-34}$ | Yes | Same |
| *TSHR* / Subcutaneous adipose | UKB hyperthyroidism | Secondary | 0.31 (0.27–0.37) | $2.86\times10^{-42}$ | Yes | Same |
| *TSHR* / Subcutaneous adipose | FinnGen GO | Secondary | 0.45 (0.32–0.63) | $5.76\times10^{-6}$ | Yes | Same |
| *TSHR* / Subcutaneous adipose | FinnGen GO strict | Secondary | 0.51 (0.35–0.74) | $3.67\times10^{-4}$ | Yes | Same |
| *TSHR* / Visceral adipose | BBJ GD | Secondary | 0.56 (0.46–0.67) | $2.64\times10^{-10}$ | Yes | Same |
| *TSHR* / Visceral adipose | FinnGen GD | Primary | 0.50 (0.43–0.58) | $4.09\times10^{-19}$ | Yes | Same |
| *TSHR* / Visceral adipose | UKB hyperthyroidism | Secondary | 0.39 (0.34–0.45) | $9.19\times10^{-38}$ | Yes | Same |
| *TSHR* / Visceral adipose | FinnGen GO | Secondary | 0.69 (0.49–0.98) | 0.0368 | No | Same |
| *TSHR* / Visceral adipose | FinnGen GO strict | Secondary | 0.78 (0.54–1.13) | 0.186 | No | Same |
| *IGF1R* / Thyroid | BBJ GD | Secondary | 1.23 (0.90–1.68) | 0.200 | No | Same |
| *IGF1R* / Thyroid | FinnGen GD | Primary | 0.76 (0.56–1.02) | 0.0692 | No | Opposite |
| *IGF1R* / Thyroid | UKB hyperthyroidism | Secondary | 1.33 (0.99–1.80) | 0.0594 | No | Same |
| *IGF1R* / Thyroid | FinnGen GO | Secondary | 1.02 (0.54–1.94) | 0.945 | No | Same |
| *IGF1R* / Thyroid | FinnGen GO strict | Secondary | 1.23 (0.62–2.44) | 0.549 | No | Same |
| *IGF1R* / Cultured fibroblasts | BBJ GD | Secondary | 0.91 (0.70–1.20) | 0.511 | No | Opposite |
| *IGF1R* / Cultured fibroblasts | FinnGen GD | Primary | 1.11 (0.89–1.38) | 0.359 | No | Same |
| *IGF1R* / Cultured fibroblasts | UKB hyperthyroidism | Secondary | 1.03 (0.82–1.29) | 0.815 | No | Same |
| *IGF1R* / Cultured fibroblasts | FinnGen GO | Secondary | 0.78 (0.49–1.25) | 0.301 | No | Opposite |
| *IGF1R* / Cultured fibroblasts | FinnGen GO strict | Secondary | 0.78 (0.47–1.30) | 0.341 | No | Opposite |

Post hoc fixed-panel extension using GTEx v8; tissue colocalization was not performed. A, availability of the ten gene–tissue combinations; F statistics are for the single retained instrument. ‘Not tested’ means the gene is not among the genes GTEx tested in that tissue. GTEx whole blood (n = 670) yielded no instrument at the fixed threshold, whereas the main analysis used eQTLGen (up to 31,684 participants). B, all 20 estimates. The primary family is the ten FinnGen GD comparisons ($P < 0.005$); the secondary family is the 40 comparisons with the other datasets ($P < 0.00125$); unestimable combinations stay in both denominators. ORs are per unit of inverse-normal-transformed tissue expression and are not comparable in size with the blood ORs. With one instrument, each *P* value equals that variant’s disease-association z-test *P* value (UKB source BOLT-LMM *P* values differ slightly). The adipose *TSHR* instruments (rs12101255, rs74064796; 50 bp apart in *TSHR* intron 1) are themselves blood *TSHR* eQTLs near rs179252 and re-express the regional association rather than providing independent evidence. Cultured fibroblasts and adipose tissue are not orbital tissue. Direction, compared with the blood estimate in the same dataset. CI, confidence interval; eQTL, expression quantitative trait locus; GTEx, Genotype-Tissue Expression; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Figure S1

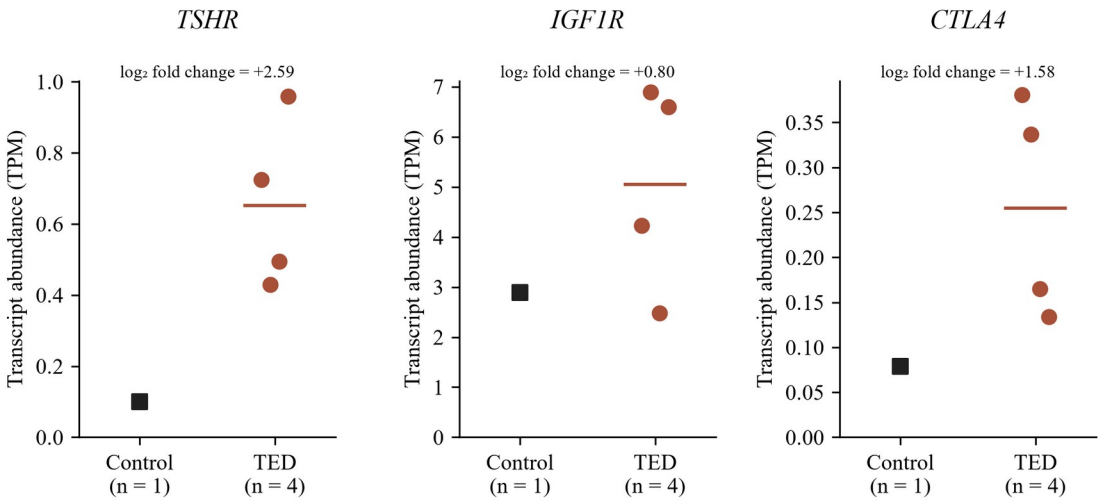



**Figure S1. Descriptive orbital expression of *TSHR*, *IGF1R* and *CTLA4*.** Transcript abundance in orbital adipose tissue collected during orbital decompression from four patients with inactive TED, and in nasal fat pad from one person without thyroid disease undergoing cosmetic upper blepharoplasty. Points are biological samples (technical replicates combined); horizontal lines show TED means. The $\log_2$ fold change is $\log_2$[(mean TED TPM + 0.01)/(control TPM + 0.01)]. Vertical axes differ by gene. The control came from a different fat depot, and one control cannot estimate control-group variance; no inferential test was performed, and these observations do not validate the genetic findings. TED, thyroid eye disease; TPM, transcripts per million.

## Figure S2

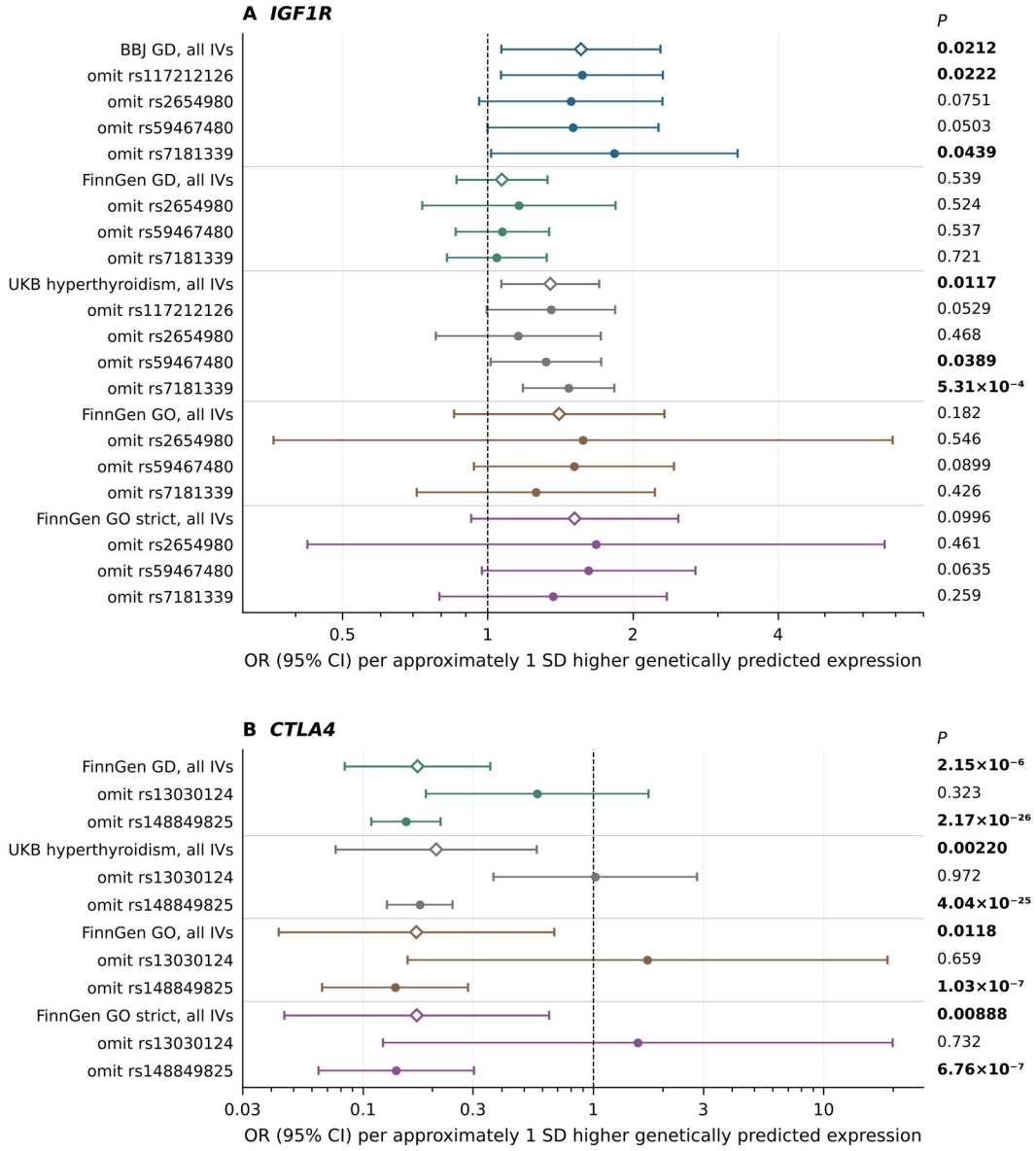



**Figure S2. Dependence of *IGF1R* and *CTLA4* estimates on individual instruments.** For each dataset, the open diamond is the estimate using all instruments (as in Table 2), and the filled points below it omit the named instrument (IVW when at least two instruments remain, Wald ratio when one remains). Lines are 95% CIs on a logarithmic axis; the dashed line is OR = 1. *P* values are shown at right; bold, nominal $P < 0.05$. (A) *IGF1R* in all five datasets. (B) *CTLA4* in the four FinnGen and UKB datasets; BBJ *CTLA4* and all *TSHR* estimates use one instrument and could not be assessed. The 25 post hoc omission estimates reuse the same association data and are not independent replications; loss of nominal significance can reflect reduced precision as well as the omitted variant's contribution. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; OR, odds ratio; SD, standard deviation; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.